Key concepts — Santélia project

A short study guide in plain English, with the French terms in brackets and examples from your own Proxmox lab.
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1. Requirements specification (cahier des charges, CDC)

A document written before building anything. It describes what the client needs, under which constraints, and how everyone will check at the end that the result is correct. It protects both sides: the client knows what they will get, the project team knows what is expected.

- **Functional requirements (exigences fonctionnelles)** — what the system must do. Example: "back up critical data automatically".
- **Non-functional requirements (exigences non fonctionnelles)** — how well it must do it: availability, performance, security, budget, deadline.
- **Constraints and risks** — budget (~15 000 €), 4-month deadline, a 2-person IT team, health-data rules (HDS).
- **Benchmark** — comparing several solutions before choosing (e.g. Proxmox Backup Server vs Bacula, Zabbix vs Nagios).
- **Acceptance tests (recette)** — the list of tests that prove the project is finished and works.

In your lab: *your Santélia CDC listed the needs (backups, monitoring, continuity), chose Proxmox Backup Server and Zabbix, and left the RTO "to be confirmed with the client". That last point is why your measured RTO could not be compared with a target.*

2. Proxmox VE clustering

Proxmox VE is a free platform for running virtual machines (VMs) and containers. A cluster links several Proxmox servers (nodes) so they can be managed together and protect each other.

- **Node (nœud)** — one Proxmox server. You had three: pve (.200), pve2 (.201), pve3 (.202).
- **Corosync** — the communication layer. Nodes constantly send each other small "I am alive" messages (heartbeats) and share cluster information. If a node stops answering, Corosync notices.
- **pmxcfs / /etc/pve** — a small shared configuration folder, copied on every node. A change made on one node appears on all of them. It becomes read-only when there is no quorum.
- **Quorum** — the majority rule. Each node has 1 vote. Quorum = (number of nodes ÷ 2, rounded down) + 1. With 3 nodes, quorum = 2. The cluster only makes changes if it has at least 2 votes.
- **Split-brain** — the danger quorum prevents: two isolated halves of a cluster both thinking they are in charge and starting the same VM twice, which can corrupt its data.
- **HA, high availability (haute disponibilité)** — Proxmox watches the VMs you give it. If their node dies, it restarts them automatically on a surviving node. Managed with ha-manager.
- **CRM and LRM** — the two parts of HA. The CRM (Cluster Resource Manager) is the "boss" that decides where each VM runs; there is one master at a time. The LRM (Local Resource Manager) is the "worker" on each node that actually starts and stops VMs.
- **Fencing** — before restarting a VM elsewhere, HA must be certain the failed node is really off. Otherwise the VM could run twice. Proxmox waits until the dead node's lock has expired (about 2 minutes).

- **Watchdog** — a timer each node must keep resetting. If a node freezes and stops resetting it, the watchdog reboots the node. This guarantees a frozen node cannot keep running a VM. In VMs, Proxmox uses a software watchdog (softdog).

In your lab: *with 2 nodes out of 3 the cluster kept working; with 1 it showed "Activity blocked". After powering off pve2, HA fenced it and restarted VM-Test on pve in 2 min 17 s. Later, a big clock jump made pve's watchdog reboot it: that was self-fencing, the same protection.*

3. Linux NFS (Network File System)

NFS lets one Linux machine share a folder over the network so other machines can use it as if it were their own disk.

- **NFS server** — the machine that owns the folder (package nfs-kernel-server).
- **/etc/exports** — the "guest list": which folder is shared, with which network, and with which rights (rw, sync, no_root_squash…).
- **exportfs -ra / exportfs -v** — apply the guest list, then show what is currently shared.
- **Why it matters for HA** — a VM can only restart on another node if that node can read its disk. Shared storage makes the disk visible to every node.
- **Limit** — if the NFS server itself fails, the shared storage disappears: it is a single point of failure (SPOF). Production uses redundant storage (NAS in HA, Ceph).

In your lab: */srv/nfs/vmstore on pve, added to the cluster as "nfs-shared" with pvesm add. Migrating VM-Test to pve2 took 6 s because only its small configuration moved, not its disk.*

4. Log-based troubleshooting

When something goes wrong, the system usually wrote down why. Reading logs and status commands is more reliable than guessing.

Command	What it tells you	When you used it
pvecm status	Cluster health: number of nodes, votes, quorum, Quorate Yes/No	Before each part; after crashes
pvecm nodes	List of cluster members	Part 2
ha-manager status	HA view: master, each LRM, where each VM runs and its state	Part 4, failure test
journalctl -u <service>	Time-stamped log messages of one service	Exact t0 / t1 / t2 of the RTO
pvesm status	Storages available on this node and their state	Checking nfs-shared on 3 nodes
qm list / qm config / qm status	VMs on this node, a VM's settings, whether it is running	Before moving the disk, after recovery
timedatectl / date / chronyc	Clock state and NTP synchronisation	Time-drift problem

Good habit: check the state, read the exact error message, change one thing, check again.

5. Business continuity and disaster recovery

	PCA — Plan de Continuité d'Activité (Business Continuity Plan)	**PRA — Plan de Reprise d'Activité (Disaster Recovery Plan)**
Goal	Keep the service running during an incident	Restart the service after a major disaster
Typical incident	One server or one disk fails	Fire in the server room, ransomware, site lost
Typical tools	Cluster, HA, redundant hardware	Backups (3-2-1 rule), off-site copy, restore procedure
Interruption	None or a few minutes	Hours, sometimes days

- **RTO, Recovery Time Objective** — the maximum acceptable time before the service works again. Answers: "How long can we be down?" Your measured RTO: 2 min 17 s.
- **RPO, Recovery Point Objective** — the maximum acceptable amount of data lost, measured in time since the last good copy. Answers: "How much work can we lose?" Daily backups mean an RPO of up to 24 h.
- **3-2-1 backup rule** — 3 copies of the data, on 2 different types of storage, with 1 copy off-site.
- **Availability** — the percentage of time a service works. 99 % ≈ 3.65 days down per year; 99.9 % ≈ 8.8 hours per year.

In your lab: *the HA cluster is the PCA side (a node failed, the service came back in about 2 minutes). Proxmox Backup Server with an off-site copy, from your CDC, is the PRA side.*